Maithil Rupesh Mehta

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Education

University of Wisconsin–Madison | GPA: 4.00/4.00 — Madison, WI
Master of Science in Computer Science — *Sep 2025 – Dec 2026*
Coursework: AI Agents, Foundation Models, Computer Vision, Machine Learning, Distributed Systems, HPC, Big Data
Teaching Assistant: Data Management for Data Science

Birla Institute of Technology and Science, Pilani | GPA: 8.29/10.00 — Pilani, India
Bachelor of Engineering in Computer Science, Minor in Data Science — *Aug 2018 – May 2022*

Experience

MathWorks — Natick, MA
Software Engineering Intern, Infrastructure Automation & Tooling — *May 2026 – Aug 2026*
- Built and productionized an **LLM agent** (Python, GPT-4o, Claude Opus) that lets non-technical users query a **50TB** infrastructure data warehouse in natural language, replacing manual API lookups across **45+** REST endpoints.
- Designed an **agent harness** around a state-machine loop (tool call, think, observe, code, respond) where the model picks each step, with an **MCP** tool layer, parameter validation, response caching, and resumable session summaries.
- Built a **200-query eval set** (50 hand-written, 150 LLM-generated and reviewed) on which the agent reached **90% task accuracy** and **95.3%** step-level trajectory match.

The D. E. Shaw Group — Hyderabad, India
Senior Member Technical (Software Engineer), Investor Relations, promoted from Member Technical — *Jul 2022 – Jun 2025*
- Led the design and build of a temporal **data platform** (Python, PostgreSQL, Kafka, React) over **100M+ records** of fund performance and risk metrics, serving **100+** internal users. Dashboard users grew **2.5**× within a month of launch and manual on-call data requests fell to near zero.
- Prototyped an **LLM-powered** (GPT-4) natural-language-to-chart tool that generated internal query-language and plotting code with sandboxed execution, and helped design eval questions for the fine-tuned query-language model.
- Cut data validation turnaround from **3–4 days to ∼30 minutes** with rule-based anomaly checks on statistics.

Amazon — Hyderabad, India
Software Engineering Intern, Finance Automation — *Feb 2022 – Jun 2022*
- Migrated **15** finance APIs from on-premises Java services to **AWS** (Lambda, S3, EC2), cutting infrastructure costs by **25%**, and reworked business logic, data models, and unit tests for the new APIs.
- Built JavaScript and Ruby on Rails tooling that flags discrepancies between vendor invoices and goods Amazon actually received (damaged or incorrect items), routing them to vendors for explanation and reconciliation.

Salesforce — Hyderabad, India
Software Engineering Intern, Big Data — *Jun 2021 – Jul 2021*
- Built a multithreaded Python service that polls **HBase** clusters across data centers for RPM deployment status, powering the production dashboard and cutting failed-deployment root-cause time by **30%**.
- Won the company-wide intern hackathon (**200+** interns) as part of a 3-person team with a computer-vision 2FA liveness check that detects facial micro-movements from screen-light reflections.

Projects

FreshLog: Freshness-Aware Context Archives for Long-Horizon Agents
- Conceived and building FreshLog (3-person team), a lossless context archive for long-horizon **LLM agents** that flags recalled tool observations made stale by later changes to files, databases, or APIs.
- Built the append-only event log and per-resource latest-state map, plus agent tools for staleness-labeled recall, history, and revalidation, evaluated on LOCA-Bench against 4 baselines for stale-decision rate, task success, and token cost.

TALMAS: Training-Free Attention Suppression for Masked Diffusion LMs [Code] [Paper]
- Conceived and designed a training-free attention intervention for masked diffusion LMs, improving **LLaDA-8B** GSM8K accuracy from **72.33% to 73.77%** and MATH Algebra from **49.6% to 51.6%** with no weight changes.

Fault-Tolerant Vector Search in Qdrant [Code] [Paper]
- Measured how replication factor trades storage for resilience in a 3-node **Qdrant** cluster (1M SIFT-128 vectors, 50 QPS) under live crashes and partitions. RF=2 matched RF=3 on crashes with zero errors and unchanged Recall@10, and RF=3's only edge was partition recovery (p99 **18.2 vs. 37.2 ms**).

Technical Skills

AI & ML: PyTorch, TensorFlow, Hugging Face, OpenAI/Anthropic APIs, MCP, RAG, LLM eval, diffusion LMs
Languages & Frameworks: Python, Java, C, C++, JavaScript, TypeScript, SQL, Ruby, React, Ruby on Rails
Data: Postgres, Spark, PyArrow, HBase, ChromaDB, Qdrant, Cassandra, Kafka, MySQL, MongoDB
Cloud & DevOps: AWS (Lambda, S3, EC2), Docker, Kubernetes, Git, Linux, Bash, gRPC, REST APIs, CI/CD